



Your Large Vision-Language Model Only Needs A Few Attention Heads For Visual Grounding

CVPR'25 Highlight

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자료조사: 이진주, 최승준, 안종식, 이찬영, 허정원

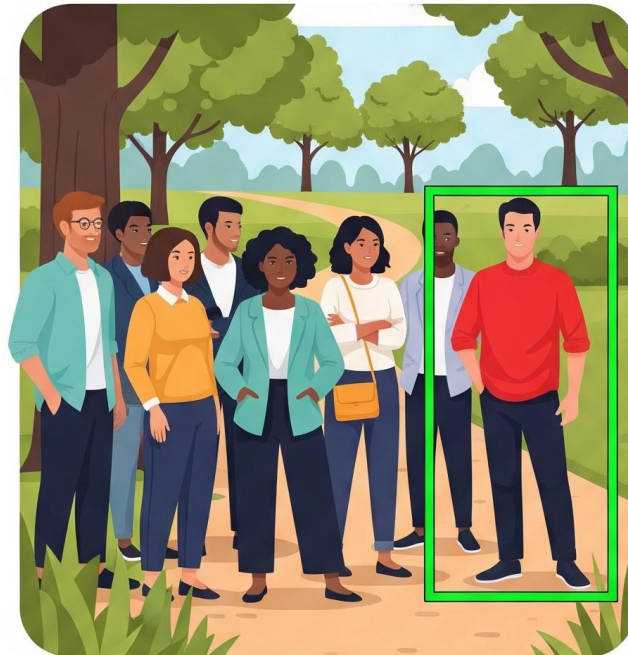
Visual Grounding

Localizing a specific region within an image based on a given text description

자유로운 텍스트 설명에 대응되는 이미지 영역을 찾는 문제

Referring Expression Comprehension/Segmentation (REC/RES)

Box mask



the man in the
red shirt

LVLM: Large Vision Language Model

구분	VLM (Vision Language Model)	LVLM (Large Vision Language Model)
주요 초점	시각(Vision)과 언어(Language)를 결합하여 이해하고 처리하는 멀티모달 능력	대규모(Large)의 시각 및 언어 데이터를 학습하여 뛰어난 일반화 및 제로샷(Zero-shot) 추론 능력을 갖춘 모델
규모	상대적으로 작거나 (초기 연구 모델), LVLM보다 일반적인 비전-언어 모델 전반.	매우 거대한 파라미터 수와 방대한 학습 데이터셋을 사용 (LLM의 'Large' 개념과 유사)
능력	이미지 캡션, 시각 질의응답 (VQA) 등 기본적인 시각-언어 연관 작업.	매우 복잡한 멀티모달 추론, 다양한 작업에 대한 뛰어난 범용성 및 상황 이해 능력.
예시	초기 VLM 연구 모델 (e.g., CLIP, ViLT), 비교적 작은 규모의 모델.	GPT-4o, Gemini, LLaVA 등 최근의 강력한 성능을 보이는 모델.

Intuition

LVLM(Large Vision-Language Model)은 텍스트-이미지 관계를 이해 → 텍스트 생성

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LVLM이 텍스트를 생성할 때 사실상 특정 이미지 영역을 이해하고 있지 않을까?

Application of LVLMs in Grounding Task

- Grounding 데이터셋으로 fine-tuning
- Localization을 위해 추가 component 붙임
- LISA: [SEG] 토큰을 마스크 임베딩으로 도입, 추가 디코더로 segmentation mask 생성
→ LVLM이 언어적 힌트를 주면, 추가 디코더가 그 힌트를 pixel단위 출력으로 바꿔 주는 구조
- F-LMM: Frozen LVLM의 Attention weight를 활용, Mask refinement 학습 필요
→ mask refinement 모듈을 visual grounding dataset으로 fine-tuning (training-free 아님)
→ 또한, 모든 attention head를 사용하여 각 head의 상대적 중요성을 고려하지 않음

Training free Visual Grounding

- CLIP-based:

Off-the-shelf 모델로 Region proposal(e.g. SAM) → CLIP 유사도로 가장 관련성 높은 box/mask 선택

Zero-shot RIS(CVPR'23), TAS(EMNLP'23)

- Diffusion-based:

텍스트-이미지 diffusion 과정의 **attention map 잔여치(residue)**를 활용해 segmentation mask 생성.

Training free Visual Grounding

- CLIP-based:

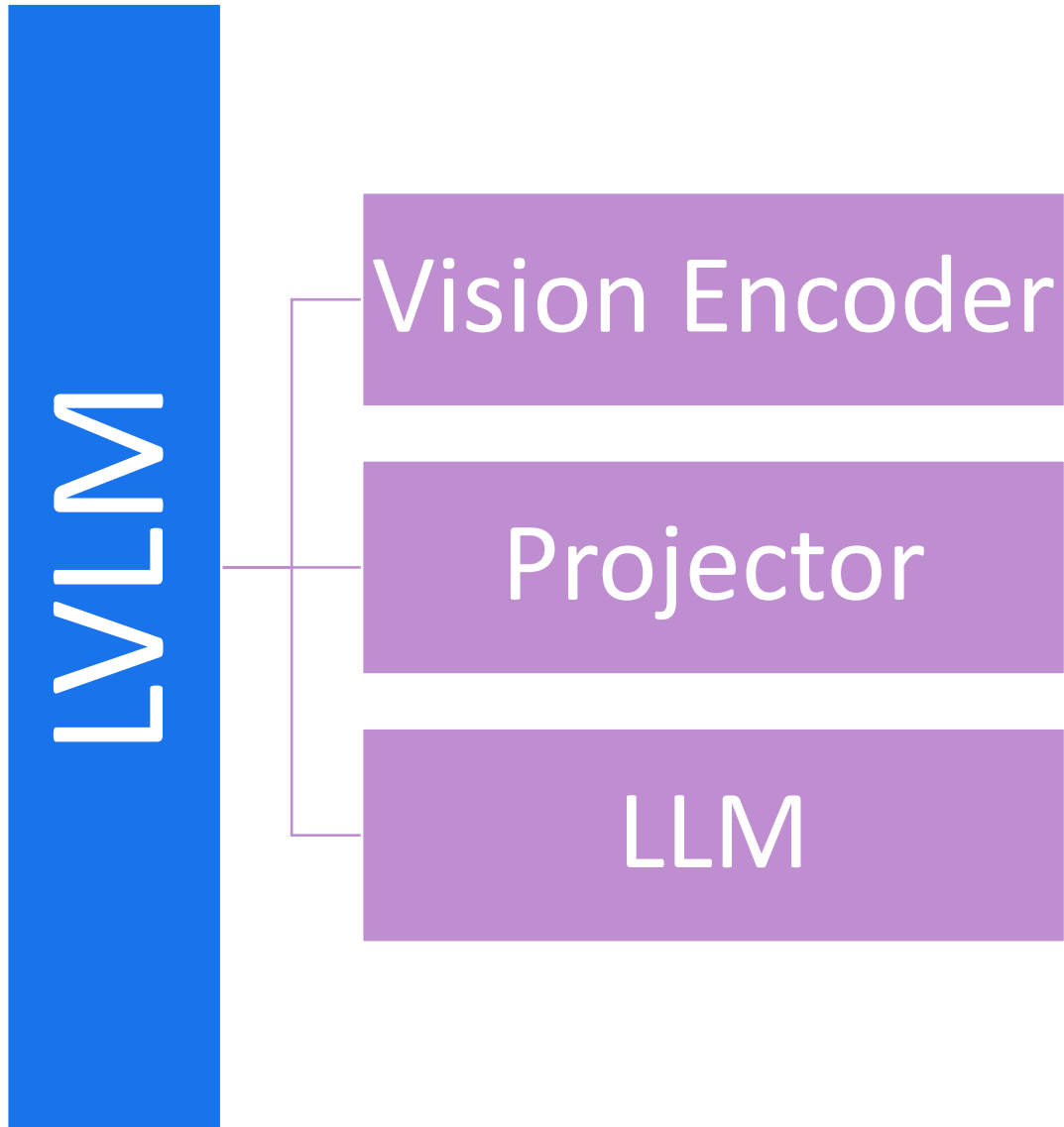
Off-the-shelf 모델로 Region proposal(e.g. SAM) → CLIP 유사도로 가장 관련성 높은 box/mask 선택
Zero-shot RIS(CVPR'23), TAS(EMNLP'23)

- Diffusion-based:

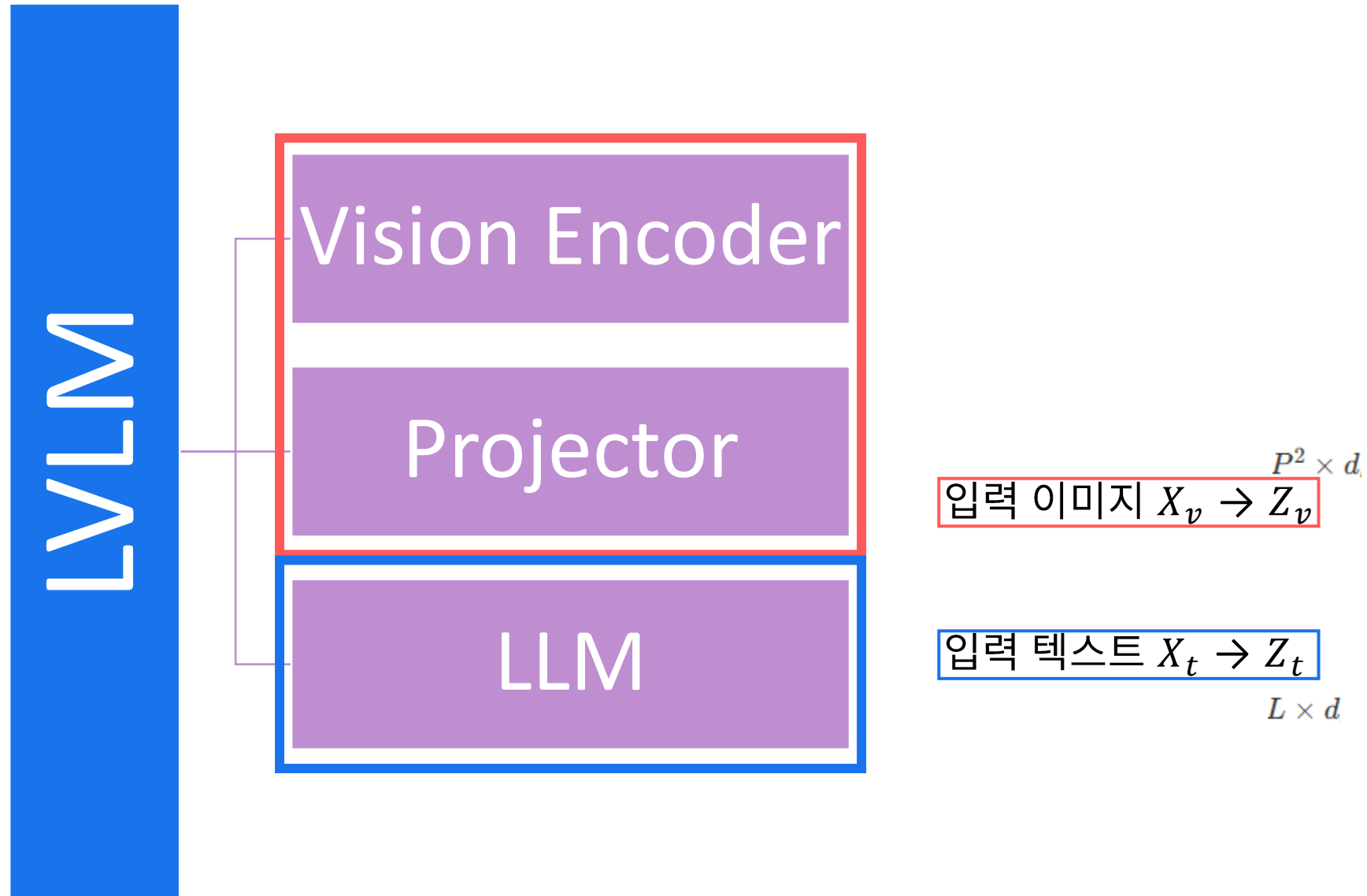
텍스트-이미지 diffusion 과정의 **attention map 잔여치(residue)**를 활용해 segmentation mask 생성.

본 논문은 LVLM 기반 최초의 Training-free Visual Grounding framework 제안

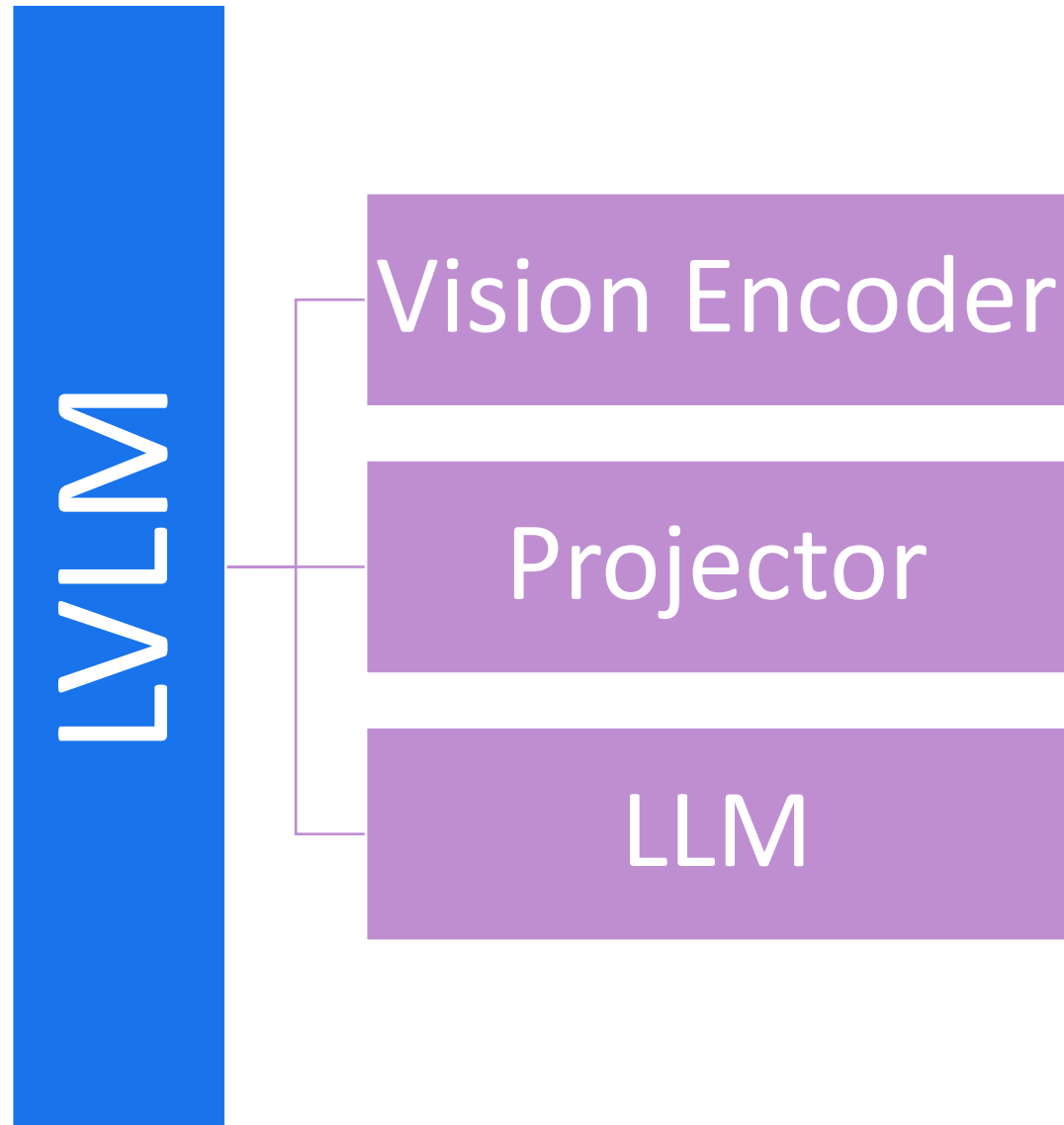
Background



Background

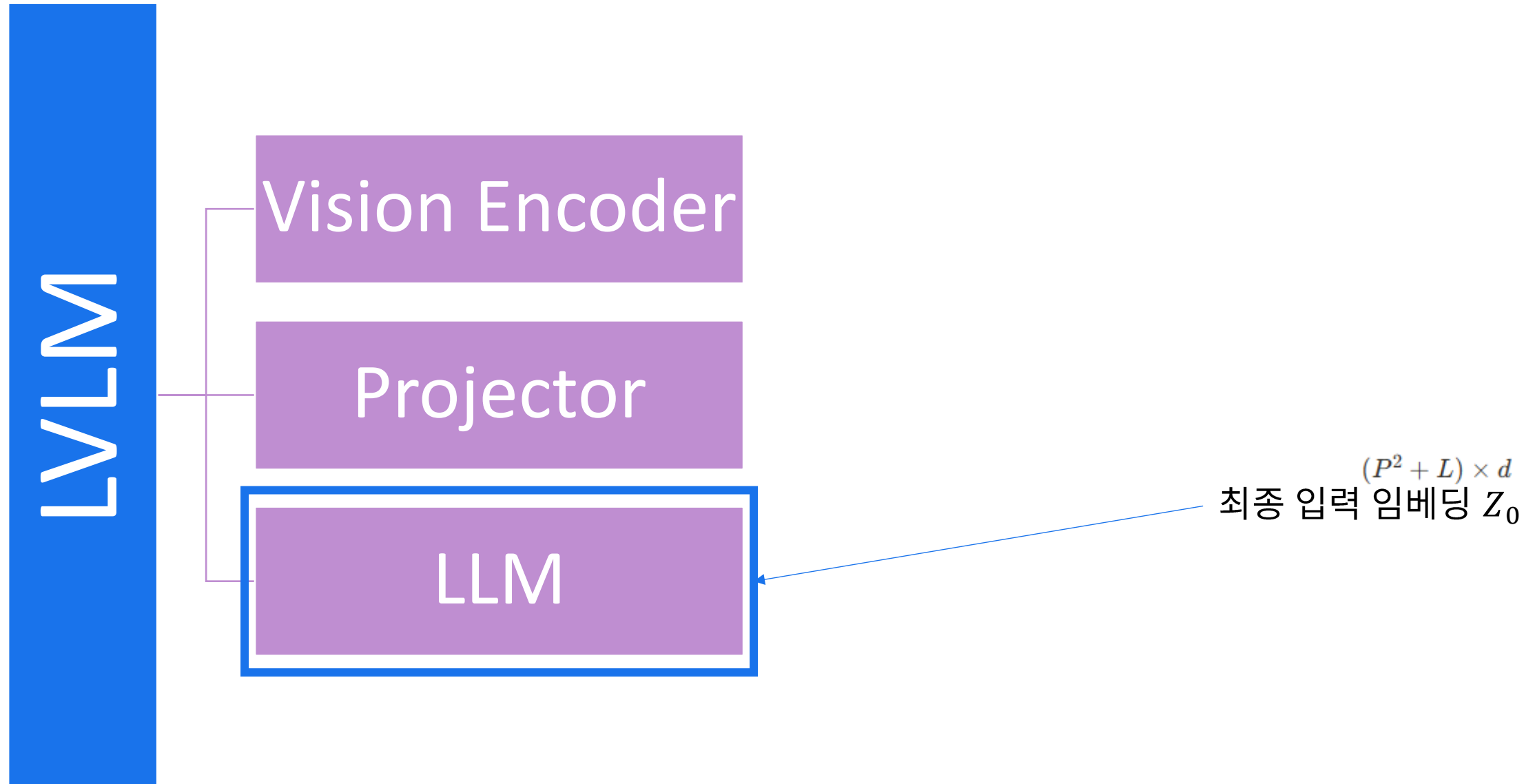


Background



$$\begin{aligned} \text{입력 이미지 } X_v &\rightarrow Z_v & Z_v &\in \mathbb{R}^{P^2 \times d} \\ \text{입력 텍스트 } X_t &\rightarrow Z_t & Z_t &\in \mathbb{R}^{L \times d} \\ & & Z_0 = [Z_v; Z_t] &\in \mathbb{R}^{(P^2+L) \times d} \end{aligned}$$

Background



Background

- 최종 입력 임베딩 Z_0 은 여러 MHSA, FFN 모듈로 이루어진 디코더 블록을 연속적으로 통과
- 토큰 간 상호작용은 Attention Head에서만 발생

$$\text{Attn}^{\ell,h}(Z^{\ell-1}) = \text{softmax} \left(\frac{QK^{\top}}{\sqrt{d_h}} \right).$$

- LLM 디코딩은 Autoregressive 방식으로 동작: 앞 토큰 정보가 뒤 토큰으로 전달
- 따라서, 문장의 마지막 텍스트 토큰을 전체 문맥의 영향을 받은 query 벡터 q_{txt} 로 설정
e.g. "the pizza mouth."에서 마지막 토큰 "."의 query 벡터를 사용함

Background

- Q를 q_{txt} , K를 이미지 토큰으로 설정하여 Attention weight 계산 → 이미지-텍스트 상호작용 분석

$$a^{\ell,h} = \text{softmax} \left(\frac{q_{\text{txt}} K^{\top}}{\sqrt{d_h}} \right) \in \mathbb{R}^{P^2+L}$$

- 분석에서는 $a^{\ell,h}[1:P^2]$ 즉 이미지 토큰에 해당하는 앞의 P^2 성분에 집중
- $L_{\ell} H_h$ 를 사용하여 층 ℓ 과 헤드 h 를 지칭. e.g. L5 H3: 5번째 층, 3번째 어텐션 헤드

How to find localization head

- Localization 특성을 가지는 Head를 어떻게 찾을 것인가?

How to find localization head

- Localization 특성을 가지는 Head를 어떻게 찾을 것인가? (2 criteria)

1. Attention Sum

2. Spatial Entropy

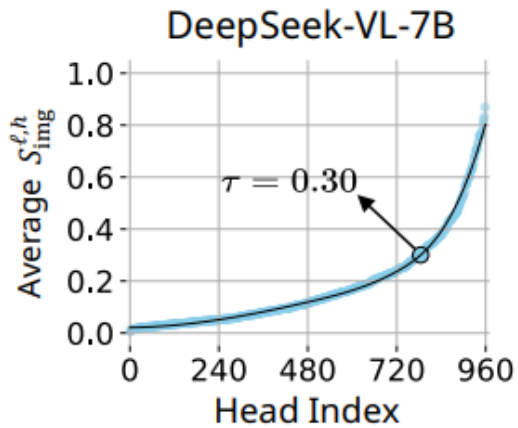
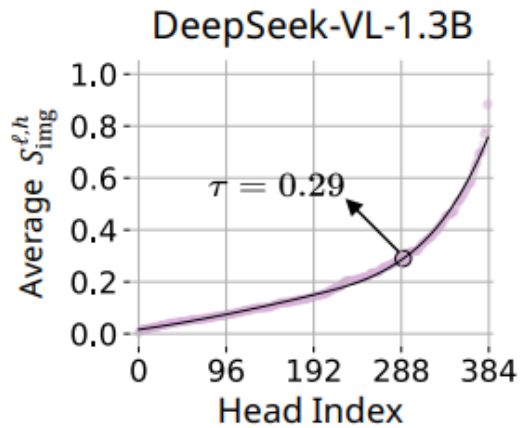
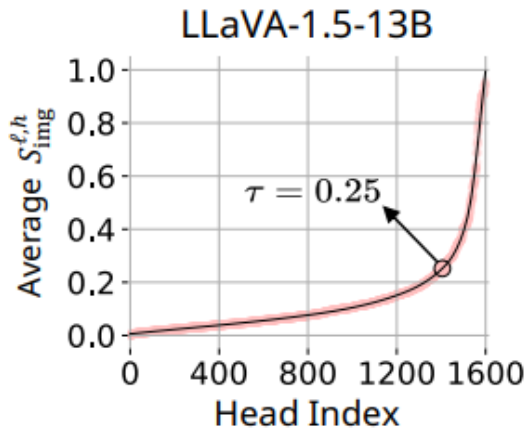
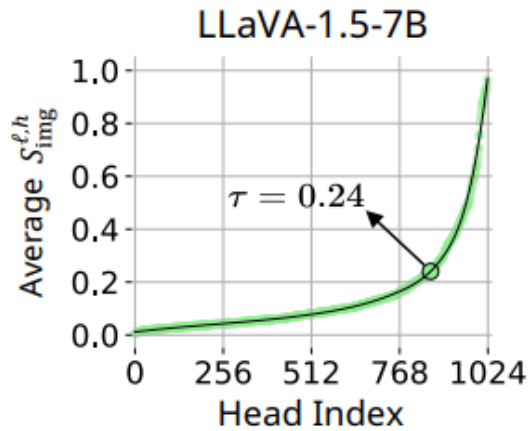
Criterion 1: Attention Sum

- RefCOCO training set에서 Random 1k 샘플을 뽑아 각 head의 평균 $S_{img}^{l,h}$ 계산

$$S_{img}^{l,h} = \sum_{i=1}^{P^2} a^{\ell,h}[i]$$

- $S_{img}^{l,h}$: 각 attention head에서 이미지 정보가 텍스트 쿼리와 얼마나 관련있는지 정량화

Criterion 1: Attention Sum



- 대부분 head는 값이 낮고, 일부 head만 텍스트-이미지 상호작용에 기여
- 높은 값/낮은 값 구분 위해 **그래프의 최대 곡률 지점**을 threshold τ 으로 설정
- $S_{img}^{l,h} \geq \tau$ 헤드가 이미지에 효과적으로 집중한다 간주

Criterion 2: Spatial Entropy

- 좋은 localization head는 높은 attention sum을 가질 뿐 아니라, 특정 객체 영역에 집중해야 함
- attention map 내에서 클러스터가 얼마나 지역적으로 형성되는지 평가 **spatial entropy**를 사용
- Spatial entropy가 낮을수록 클러스터가 잘 형성되어 있음

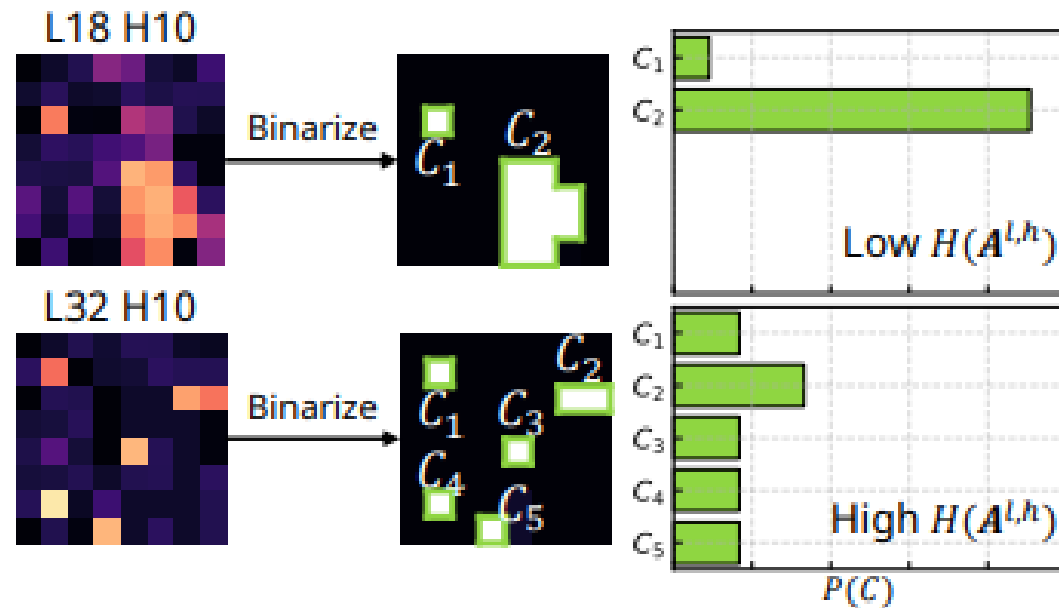


Figure 4. Illustration of the process for calculating spatial entropy. The attention map is binarized, and the spatial entropy is computed based on the sizes of its connected components $\{C_i\}_{i=1}^N$.

Criterion 2: Spatial Entropy

Spatial entropy

1. $a^{l,h}[1:P^2]$ 을 $P \times P$ 맵 $A^{l,h}$ 로 reshape
2. 평균값 이상인 위치=1, 이하=0으로 이진화
3. 8-이웃 연결 요소(connected component) 집합 $\{C_i\}_{i=1}^N$ 추출 (클러스터링)
4. 각 클러스터의 비율 계산 $P(C_i) = |C_i| / \sum_{i=1}^N |C_i|$
5. Shannon entropy로 계산:

$$H(A^{l,h}) = - \sum_{i=1}^N P(C_i) \log P(C_i).$$

0	0	0	0	1	0
0	1	0	1	1	1
0	1	1	0	0	1
1	0	1	1	1	0
1	1	0	1	0	0

4-connectivity map

0	0	0	0	1	0
0	1	0	1	1	1
0	1	1	0	0	1
1	0	1	1	1	0
1	1	0	1	0	0

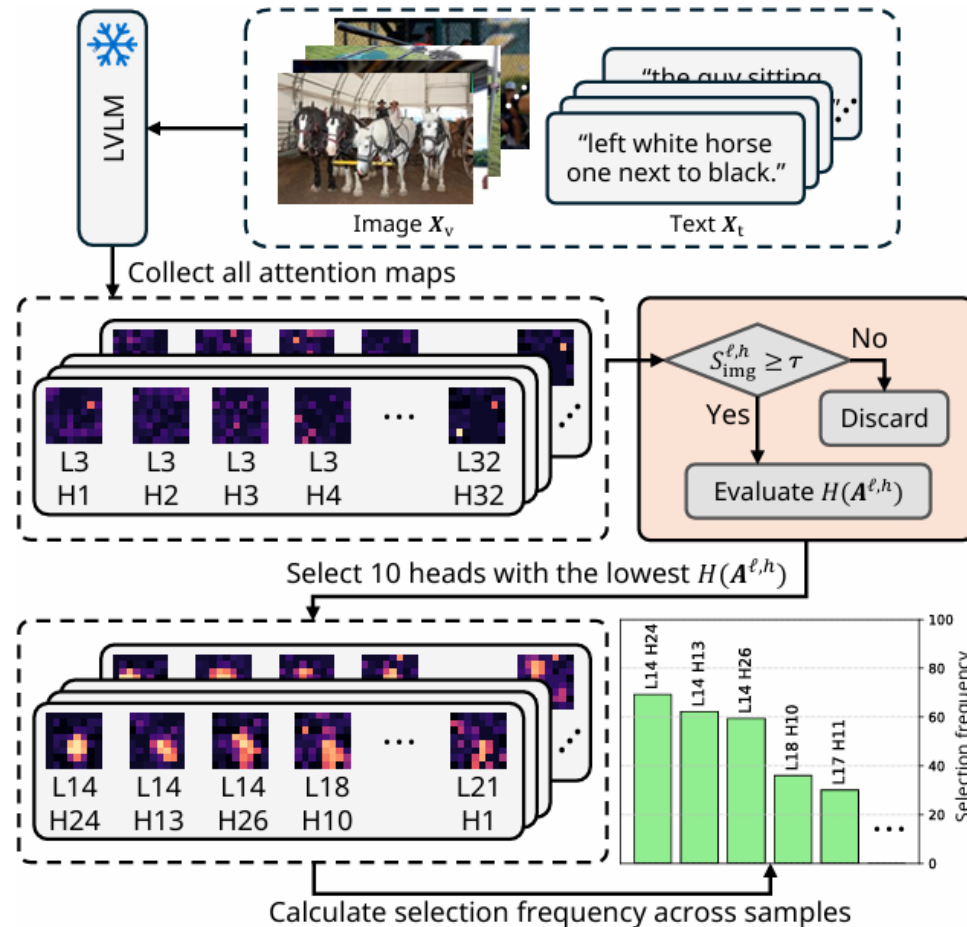
8-connectivity map

Finding Localization Heads via Criteria

Criterion 1

Criterion 2

- Attention sum 기준으로 head filtering, 이 중 10-lowest spatial entropy인 헤드 뽑음
- 특정 head가 얼마나 자주 포함되는지 빈도 계산 → selection frequency

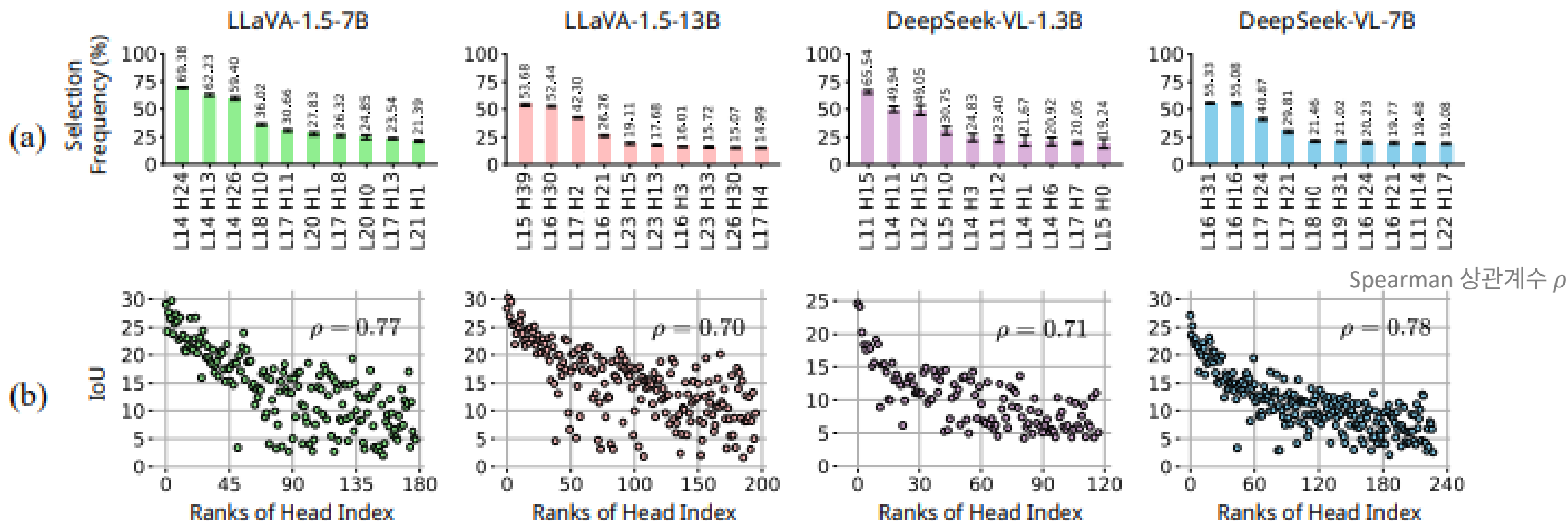


Finding Localization Heads via Criteria

Criterion 1

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- Attention sum 기준으로 head filtering, 이 중 10-lowest spatial entropy인 헤드 뽑음
- 특정 head가 얼마나 자주 포함되는지 빈도 계산 → selection frequency (a)
- 각 head의 attention map을 이진화하여 pseudo mask 생성, GT 사이의 IoU 계산 (b)



Localization Head

- Selection frequency에 기반해 head를 순위화, 상위 $k(3)$ 개의 head 선택 → **Localization head**
- Localization head에 Gaussian smoothing 적용하여 랜덤 노이즈 최소화
- 생성된 map들을 element-wise summation으로 합쳐서 combined map 얻음
- combined map을 이진화하여 pseudo-mask를 생성

Convex Hull algorithm

- Bbox: Pseudo mask를 둘러싸는 가장 큰 직사각형
- Mask: bbox를 SAM 프롬프트로 활용

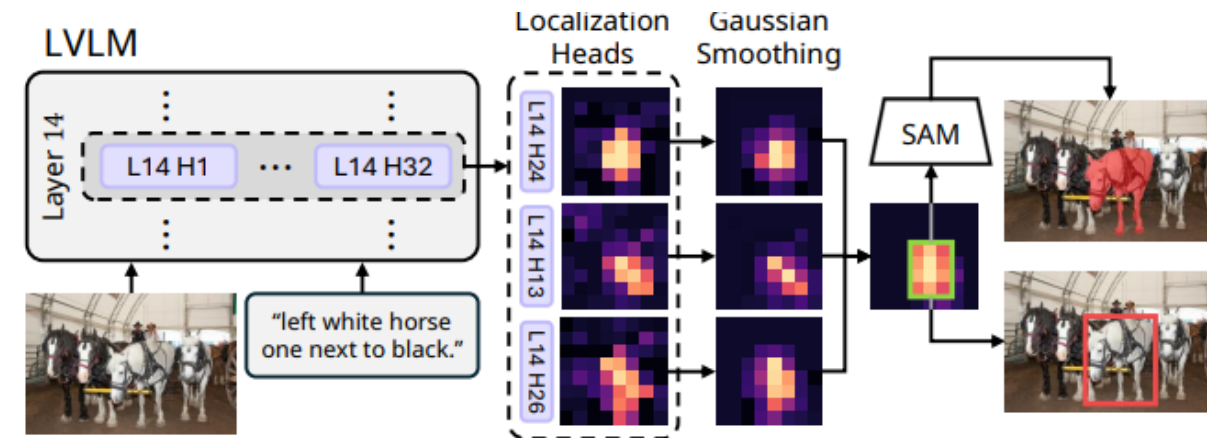


Figure 7. Our training-free visual grounding framework. Attention maps of localization heads are assembled into a combined map, which is then used to define the bounding box or segmentation mask.

Experiments

모델

- **10개의 LVLM**(예: DeepSeek-VL, Mini-Gemini, InternVL, Yi-VL, ShareGPT4V, LLaVA, LLaVA-1.5)
- 모델 크기는 1.3B ~ 13B 파라미터 규모.
- Localization head는 **항상 k=3**으로 고정.

벤치마크

- **REC(Acc@0.5)/RES(cIoU)**
- 데이터셋: RefCOCO, RefCOCO+, RefCOCOg
- ReasonSeg (더 복잡한 reasoning과 world knowledge 필요) (**cIoU**)

Main Results

Table 1. Comparison of our method with existing fine-tuning based and training-free methods on the REC (Referring Expression Comprehension) task. All fine-tuning based methods are trained on the training set of the corresponding datasets. Best performance is colored in red for fine-tuning and in blue for training-free methods.

Method	RefCOCO			RefCOCO+			RefCOCOg	
	val	testA	testB	val	testA	testB	val	test
<i>Fine-tuning based methods</i>								
MDETR [21]	86.8	89.6	81.4	79.5	84.1	70.6	81.6	80.9
SeqTR [77]	87.0	90.2	83.6	78.7	84.5	71.9	82.7	83.4
G-DINO [37]	89.2	91.9	86.0	81.1	87.4	74.7	84.2	84.9
ONE-PEACE [60]	92.6	94.2	89.3	88.8	92.2	83.2	89.2	89.3
UNINEXT [31]	92.6	94.3	91.5	85.2	89.6	79.8	88.7	89.4
<i>Fine-tuning based methods w/ LVLMs</i>								
Shikra-7B [6]	87.0	90.6	80.2	81.6	87.4	72.1	82.3	82.2
Ferret-7B [68]	87.5	91.4	82.5	80.8	87.4	73.1	83.9	84.8
Shikra-13B [6]	87.8	91.1	81.8	82.9	87.8	74.4	82.6	83.2
Ferret-13B [68]	89.5	92.4	84.4	82.8	88.1	75.2	85.8	86.3
CogVLM-17B [61]	92.8	94.8	89.0	88.7	92.9	83.4	89.8	90.8
<i>Training-free methods</i>								
ReCLIP [52]	45.8	46.1	47.1	47.9	50.1	45.1	59.3	59.0
Han et al. [15]	49.4	47.8	51.7	48.9	50.0	46.9	61.0	60.0
GroundVLP [48]	65.0	73.5	55.0	68.8	78.1	57.3	74.7	75.0
<i>Training-free methods w/ LVLMs (Ours)</i>								
DeepSeek-VL-1.3B	73.2	77.7	70.7	62.0	66.7	57.1	65.2	69.3
Mini-Gemini-2B	74.0	77.5	71.1	62.5	67.8	59.3	65.1	69.3
InternVL-6B	85.2	86.4	78.5	78.0	83.3	71.9	81.1	80.5
Yi-VL-6B	85.1	86.8	78.4	78.9	84.2	72.2	80.5	80.9
DeepSeek-VL-7B	85.3	87.2	81.0	77.8	83.9	73.5	81.1	82.8
ShareGPT4V-7B	86.1	87.1	80.5	79.7	86.2	71.3	82.4	82.9
LLaVA-7B	80.3	83.5	77.4	74.5	80.2	69.3	77.5	77.1
LLaVA-1.5-7B	86.5	89.8	80.2	80.1	86.3	71.9	82.3	83.0
LLaVA-13B	82.8	85.3	79.8	79.3	82.4	73.0	79.8	79.5
LLaVA-1.5-13B	87.2	90.0	83.3	82.7	88.5	74.0	84.3	85.5

Table 2. Comparison of our method with existing fine-tuning based and training-free methods on the RES (Referring Expression Segmentation) task. All fine-tuning based methods, except for LISA [26] and GSVA [65], are trained on the training set of the corresponding datasets. Red and blue colors are used as in Tab. 1.

Method	RefCOCO			RefCOCO+			RefCOCOg	
	val	testA	testB	val	testA	testB	val	test
<i>Fine-tuning based methods</i>								
LAVT [67]	72.7	75.8	68.8	62.1	68.4	55.1	61.2	62.1
ReLA [34]	73.8	76.5	70.2	66.0	71.0	57.7	65.0	66.0
UniRef++ [63]	79.1	82.1	77.5	68.4	74.0	61.5	71.4	72.8
UNINEXT [31]	82.2	83.4	81.3	72.5	76.4	66.2	74.4	76.4
<i>Fine-tuning based methods w/ LVLMs</i>								
LISA-7B [26]	74.1	76.5	71.1	62.4	67.4	56.5	66.4	68.5
GSVA-7B [65]	76.4	77.4	72.8	64.5	67.7	58.6	71.1	72.0
LISA-13B [65]	73.4	76.2	69.5	62.3	66.6	56.3	68.2	68.5
GSVA-13B [65]	77.7	79.9	74.2	68.0	71.5	61.5	73.2	73.9
GLaMM [45]	79.5	83.2	76.9	75.9	78.7	68.8	76.8	78.4
PSALM [74]	83.6	84.7	81.6	72.9	75.5	70.1	73.8	74.4
<i>Training-free methods</i>								
Yu et al. [71]	24.9	23.6	24.7	26.2	24.9	25.8	31.1	31.0
TAS [53]	29.5	30.3	28.2	33.2	38.8	28.0	35.8	36.2
Ref-Diff [40]	35.2	37.4	34.5	35.6	38.7	31.4	38.6	37.5
<i>Training-free methods w/ LVLMs (Ours)</i>								
DeepSeek-VL-1.3B	56.3	57.0	52.7	51.2	55.5	49.2	52.3	55.8
Mini-Gemini-2B	59.8	60.3	55.5	56.3	59.9	51.8	55.1	60.3
InternVL-6B	62.1	65.8	60.9	62.2	65.5	55.5	63.5	65.4
Yi-VL-6B	62.5	65.8	60.7	61.0	65.3	56.0	64.0	67.0
DeepSeek-VL-7B	73.9	76.6	70.7	63.1	66.1	56.5	64.0	68.9
ShareGPT4V-7B	73.5	76.7	70.1	59.4	63.8	55.9	60.7	65.1
LLaVA-7B	65.4	66.2	61.1	59.9	63.2	52.7	59.7	63.3
LLaVA-1.5-7B	74.2	76.5	70.4	62.5	65.2	56.0	64.2	68.1
LLaVA-13B	66.8	68.0	63.7	62.3	66.9	57.3	65.0	68.2
LLaVA-1.5-13B	76.1	78.9	72.8	64.1	67.1	57.3	67.7	69.0

Main Results

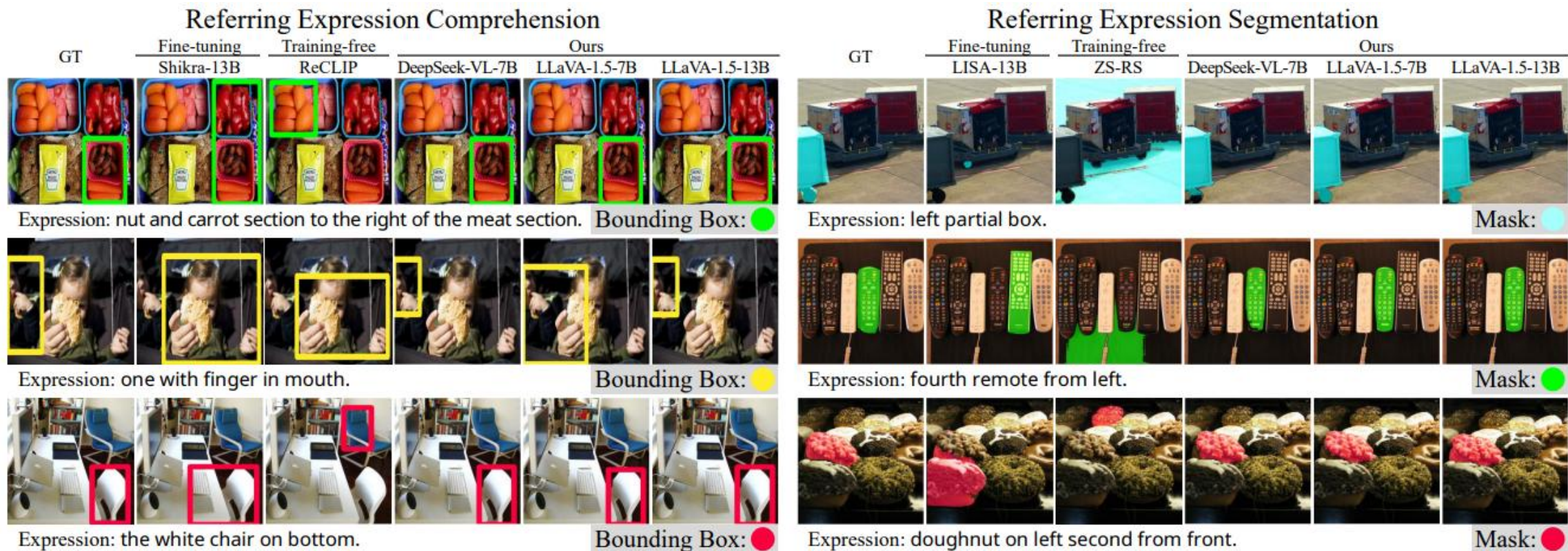


Figure 8. Qualitative results of our framework with the baseline models. LVLMs successfully localize the referred objects in various challenging scenarios including multiple similar objects, non-salient objects, and complex spatial relations.

Main Results

Table 3. Comparison of our framework with LISA [26] on the ReasonSeg (Reasoning Segmentation) benchmark.

Method	val	test		
	overall	short	long	overall
LISA-7B [26]	52.3	48.5	48.9	48.8
LISA-13B [26]	60.3	50.0	50.9	50.8
LLaVA-1.5-7B (Ours)	52.4	48.0	49.1	48.7
LLaVA-1.5-13B (Ours)	60.5	48.7	51.0	49.9

Ablations: # of Localization heads

Table 4. Ablation study on the number of localization heads (k) on the RefCOCO validation set for the RES task.

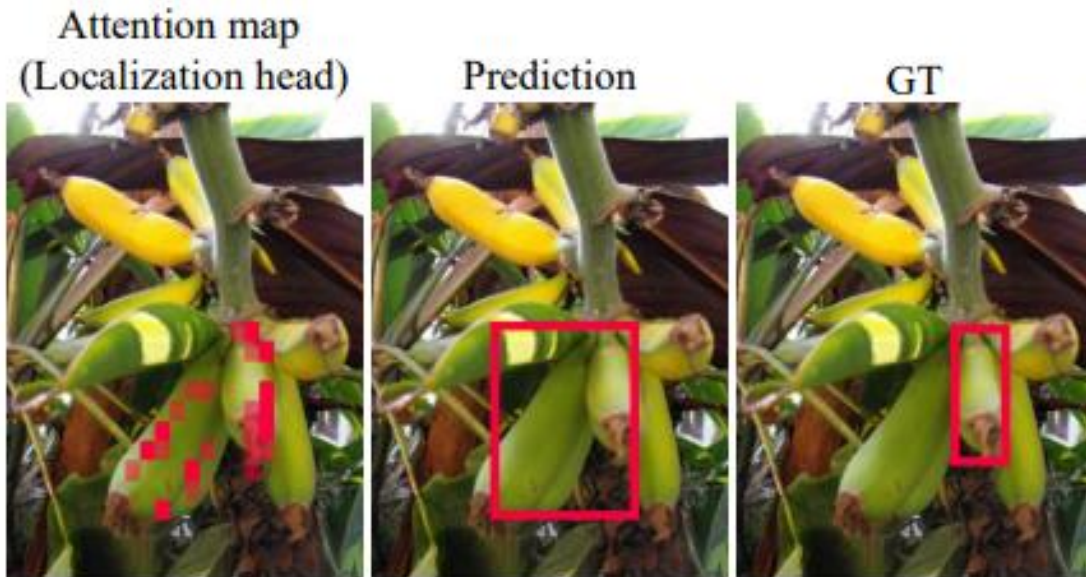
Method	k (# of Localization Heads)				
	1	2	3	4	5
DeepSeek-VL-1.3B	55.1	56.3	56.3	55.3	51.2
MiniGemini-2B	58.0	58.5	59.8	59.1	54.2
InternVL-6B	61.3	61.8	62.1	61.0	55.7
Yi-VL-6B	61.8	62.1	62.5	62.6	55.4
DeepSeek-VL-7B	70.1	72.2	73.9	73.0	65.3
ShareGPT4V-7B	70.3	72.4	73.5	73.5	60.8
LLaVA-7B	62.7	63.1	65.4	65.3	57.7
LLaVA-1.5-7B	70.3	73.1	74.2	74.1	65.4
LLaVA-13B	63.5	64.7	66.8	66.4	57.8
LLaVA-1.5-13B	71.7	75.7	76.1	76.0	65.7
Average	64.5	66.0	67.1	65.4	58.9

Ablations: Criteria

Table 5. Ablation study on the validation of criteria and selection methods for localization heads. The results are reported on the Re-fCOCO validation set and LLaVA-1.5-13B.

Criteria		Selection		REC	RES
$S_{\text{img}}^{\ell,h}$	$H(\mathbf{A}^{\ell,h})$	Fixed	Greedy		
✓			✓	23.7	18.8
	✓		✓	29.8	21.5
✓	✓		✓	67.4	63.8
✓		✓		23.9	19.3
	✓	✓		31.3	25.7
✓	✓	✓		87.2	76.1

Failure case



Expression: third from right.

Figure 9. Failure case of the LLaVA-1.5-13B [35] in visual grounding. The text-to-image attention map from a localization head (L15 H39) shows where the model focuses, helping to understand the model’s failure.

“The third banana from the right”

Multi-Object Grounding Tasks



Figure 10. Multi-object segmentation results from the localization heads of DeepSeekVL-7B, along with the corresponding raw attention maps.

Table 9. Performance comparison with F-LMM [64] on the PNG [12] benchmark.

<i>PNG (all)</i>	Ours	F-LMM
DeepSeekVL-7B	66.7	65.7

Conclusion

- LVLN 내부에 존재하는 특정 attention head (= localization head)를 발견
- Training-free visual grounding 프레임워크를 제안
- fine-tuning 기반 방법들과 비교해도 경쟁적 수준 달성
- Attention 메커니즘의 고유 성질만으로도 텍스트 기반 object localization 수행 가능성 입증